

Inventory Management & Reporting System

This project is built using:

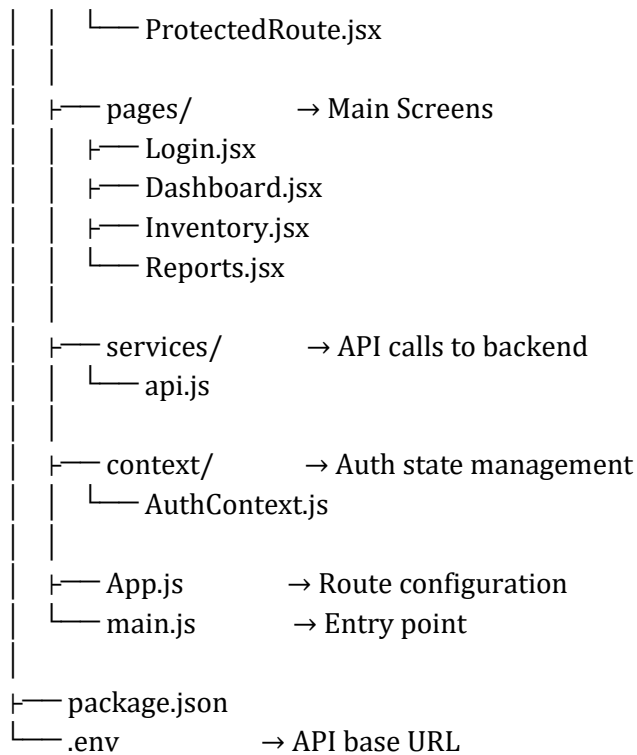
- **Frontend:** React
- **Backend:** Spring Boot
- **Database:** MySQL (or any relational database)

The system is divided into two main parts:

Frontend (React) → Backend (Spring Boot) → Database

Project Structure

```
inventory-system/
├── backend/                → Spring Boot Application
│   ├── src/main/java/com/inventory
│   │   ├── config/        → Security & configuration
│   │   ├── controller/    → REST APIs
│   │   ├── service/       → Business logic
│   │   ├── repository/    → Database access
│   │   ├── model/         → Entity classes
│   │   ├── dto/           → Request & Response objects
│   │   ├── exception/     → Error handling
│   │   └── InventoryApplication.java
│   ├── src/main/resources/
│   │   └── application.properties
│   └── pom.xml
└── frontend/              → React Application
    ├── public/            → Static files
    │   └── index.html
    └── src/
        ├── components/    → Reusable UI parts
        │   ├── Navbar.jsx
        │   └── Sidebar.jsx
```



Explanation

Backend: Handles business logic, database interaction, and REST APIs.

Frontend: Handles user interface and communicates with backend APIs.

Config Folder: Contains security and configuration settings.

Controller Folder: Manages API endpoints.

Service Folder: Contains business logic.

Repository Folder: Handles database operations.

Model Folder: Contains entity classes mapped to database tables.

DTO Folder: Defines request and response objects.

Exception Folder: Handles global error handling.

Components Folder (Frontend): Reusable UI components.

Pages Folder (Frontend): Main application screens.

Services Folder (Frontend): Handles API communication.

Context Folder (Frontend): Manages authentication state.